

Han Chen

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EXPERIENCE

Arch Jun. 2025 - Present

Operations Engineer | Software Engineering & Automation

- Built and maintained web-scraping infrastructure using Axios, Selenium, and CDP Agents across 300+ GP portals, automating authentication, navigation, and document retrieval for 10,000+ documents per day, representing up to 90% of all documents processed.
- Developed and maintained document parsing tools using Regex, OCR, and LLMs to classify, extract, and reference check 8,000+ documents per day, representing up to 75% of all documents processed.
- Developed an AI training tool that enables custom instructions at the investment, issuing-entity, and firm levels, achieving 90%+ extraction/classification accuracy while allowing operations teams to customize processing behavior without engineering changes.
- Designed the database schema and built a reporting-cadence system tracking 10,000+ funds, providing a centralized source of truth for data-quality checks, automated population, and alerting, reducing data discrepancies by 10%+.
- Improved page latency by up to 40% through frontend profiling, React memoization, bulk database operations, query optimization, and parallelized requests.

AECOM Tishman Aug. 2024 - May 2025

Senior Software Engineer (Full-Time Contract)

- Built a Python-based data extraction system using Jinja2, Pyppeteer, openpyxl, and REST APIs to automate extraction of 300+ GB of project document data per project from Procore.
- Developed an automated reporting pipeline using Python, pandas, NumPy, Azure Functions, and Power Automate to generate and distribute weekly reports containing key project financial metrics.
- Improved system API performance by 70%+ through parallel processing and data caching, while optimizing SQL queries to eliminate unnecessary database operations and reduce infrastructure costs by 20%+.

Port Authority of NY NJ Feb. 2023 – Aug. 2024

Full Stack Software Engineer (Full-Time Contract)

- Migrated a legacy .NET Core backend from v3.1 to .NET 8, resolving 150+ compatibility issues, improving system load times by 30%+, and upgrading security protocols to modern standards.
- Built an ERP system for job management and budget reporting using SPFx and TypeScript, automating job tracking and reporting and increasing job allocation efficiency by 15%+.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, C#, Ruby, SQL, HTML, CSS

Frameworks & Libraries: React, Angular, Next.js, .NET, Ruby on Rails, Node.js, Express.js, Django, Redux, Tailwind

Technologies: AWS, Azure, REST APIs, CI/CD, Azure DevOps, Git, SQL Server, PostgreSQL, MongoDB, Selenium, Pandas, NumPy, Webpack, Power BI

PROJECTS

hanchen.dev – web-responsive portfolio site – Next.js, TypeScript, Tailwind, Framer Motion

[Live Site](#) | [GitHub](#)

cozybnb – Full stack web-responsive application

[GitHub](#)

- Full-stack Airbnb clone built with React, Redux, Ruby on Rails, AWS S3, and Google Maps APIs, implementing CRUD functionality and interactive UI components.

EDUCATION

New York University - Tandon School of Engineering B.A.S in Business and Technology Management

New York, NY